Gromov Compactness for Genus 0 Pseudo-Holomorphic Curves

A Formalization Blueprint in Lean 4

Autoformalization Team

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Chapter 1

Status and conventions

The master theorem Theorem 6.7 is a machine-checked *reduction*: it is proved in Lean with no `sorry`, from explicit hypotheses called certificates (Chapter 3). All library declarations depend only on `propext`, `Classical.choice` and `Quot.sound` (census: `tests/PhaseAxiomCensus.lean`). Certificates were audited for satisfiability in `tests/CertificateSanity.lean`; two earlier certificates were shown to be too strong and replaced (Chapter 8). Several certificates are close to, or identical to, the conclusion of the theorem they serve; this is recorded in each node.

Chapter 2

Foundations

Definition 2.1 (Closed tame target)**.** A compact boundaryless smooth manifold M with pointwise nondegenerate skew forms ω_x, closed in charts, an almost-complex structure J with $J^2 = -1$, smooth in charts, and tameness $\omega_x(v, J_x v) > 0$ for $v \neq 0$.

Definition 2.2 (Compatible metric)**.** $g_x(v, w) = \frac{1}{2}(\omega_x(v, J_x w) + \omega_x(w, J_x v))$.

Lemma 2.3 (Metric properties)**.** *g is symmetric, positive definite and J-invariant.*

Proof. Algebra from tameness and $J^2 = -1$. □

Definition 2.4 (J-holomorphic map)**.** u is smooth on s and $du \circ j = J \circ du$ at every point of s.

Lemma 2.5 (Reparametrization invariance)**.** *J-holomorphicity is preserved by biholomorphic reparametrization of the domain.*

Proof. Chain rule for `mfderiv`. □

Definition 2.6 (Möbius frames)**.** Möbius transformations of $\mathbb{CP}^1$, with projective compatibility (`is_moebius`), composition, inverse, and centered stereographic affine rescalings $w \mapsto a + \lambda w$ (`affine_scaling`).

Theorem 2.7 (Three-point transitivity)**.** *Any triple of distinct points of $\mathbb{CP}^1$ can be sent to any other by a unique Möbius map.*

Proof. Adjugate matrices. □

Definition 2.8 (Energy measure and total energy)**.** The push-forward from the stereographic chart of Lebesgue measure with density $\frac{1}{2}|du|_g^2 \cdot 4/(1 + |w|^2)^2$; $E(u)$ is its real total.

Theorem 2.9 (Conformal invariance of energy)**.** *Energy is invariant under Möbius reparametrization.*

Proof. Change of variables with the spherical Jacobian. □

Lemma 2.10 (A nonconstant sphere of finite positive energy)**.** $\mathrm{id} : \mathbb{CP}^1 \to \mathbb{CP}^1$ *is J-holomorphic with* $0 < E(\mathrm{id}) < \infty$.

Proof. Energy density of the identity is identically 1. □

Definition 2.11 (Smooth local convergence)**.** C^∞_{loc} convergence on an open subset of $\mathbb{CP}^1$.

Lemma 2.12 (Closedness of J-holomorphicity)**.** C^∞_{loc} *limits of J-holomorphic maps are J-holomorphic.*

Proof. Pass to the limit in $du \circ j = J \circ du$. □

Lemma 2.13 (Diagonal extraction)**.** *A nested sequence of subsequences has a diagonal subsequence eventually inside each.*

Proof. Cantor diagonal argument. □

Definition 2.14 (Stable sphere tree limit and compactness witness)**.** A finite tree of J-holomorphic spheres with nodal matching and stable ghosts; a witness records a subsequence, frames, C^∞_{loc} convergence, frame collapse and energy convergence.

Theorem 2.15 (Component bounds)**.** $Ph \le B$*; for* $|V| > 1$*,* $G \le P - 2$ *and* $|V| \le 2P - 2$*;* $|V| \le \max(1, 2\lfloor B/h \rfloor - 2)$.

Proof. Handshake lemma and the energy gap packaged in `TargetThresholds`. □

Chapter 3

Certificates (assumed hypotheses)

Theorem 3.1 (Finite extended energy (`h_fin` discharged))**.** *Every J-holomorphic sphere $u : \mathbb{CP}^1 \to M$ (and every C^1 map on a `BoundedChartedTarget`) has finite extended energy $\mu_u(\mathbb{CP}^1) < \top$, discharging hypothesis `h_fin` (Theorem 9.2).*

Proof. For any J-holomorphic sphere under a sphere energy gap $\varepsilon_{\mathrm{sph}} > 0$, constant maps have $\mu_u(\mathbb{CP}^1) = 0 < \top$ while nonconstant maps satisfy $0 < \varepsilon_{\mathrm{sph}} \le E(u) = (\mu_u(\mathbb{CP}^1))$.toReal, forcing $\mu_u(\mathbb{CP}^1) < \top$; independently, on any `BoundedChartedTarget`, every C^1 map is globally Lipschitz on compact $\mathbb{CP}^1$ and has uniformly bounded Riemannian energy density (Theorem 9.2). □

Definition 3.2 (Universal thresholds)**.** A bundle of constants $\varepsilon_{\mathrm{reg}}, C_{\mathrm{reg}}, \varepsilon_{\mathrm{sph}}, \varepsilon_{\mathrm{neck}}, C_{\mathrm{neck}}, a, h$ together with the estimates they satisfy (ε-regularity, energy gap, neck decay).

Theorem 3.3 (Universal bubbling thresholds (`TargetThresholds` construction))**.** *On any closed tame target satisfying the proved geometric properties Theorems 3.4, 3.5 and 3.7 (or `UniformNormalChartCerti` on the canonical normal-chart family `construct_uniform_normal_chart_family` from Theorem 9.3, or first-principles normal-chart Weitzenböck, radial monotonicity and collar Widman hole-filling bounds on a `BoundedChartedTarget`, or unconditionally on discrete, subsingleton, or symplectically aspherical targets), `TargetThresholds` is constructively inhabited with universal bubbling quantum $0 < h \le \min(\varepsilon_{\mathrm{reg}}/2, \varepsilon_{\mathrm{sph}}, \varepsilon_{\mathrm{neck}}/2)$ and $\varepsilon_{\mathrm{sph}} \le \varepsilon_{\mathrm{reg}}$.*

Proof. Combining ε-regularity (`epsilon_regularity` from Theorem 3.4), the sphere energy gap (`sphere_energy_gap` from Theorem 3.5), and two-sided cylindrical energy decay (`cylinder_exponential_decay` from Theorem 3.7), `make_target_thresholds` sets $\varepsilon'_{\mathrm{sph}} = \min(\varepsilon_{\mathrm{sph}}, \varepsilon_{\mathrm{reg}}) > 0$ and $h = \min(\varepsilon_{\mathrm{reg}}/2, \min(\varepsilon'_{\mathrm{sph}}, \varepsilon_{\mathrm{neck}}$ 0 (`TargetThresholds.of_properties`). On a `BoundedChartedTarget`, metric coercivity (`exists_metric_coer` and annular extended-energy finiteness (`energyMeasureExtended_ne_top_of_jhol_uncond`) are

discharged unconditionally in `TargetThresholds.of_bounded_geometry_first_principles`, normal-chart certificates feed `TargetThresholds.of_bounded_geometry_certs` via `construct_uniform_normal_chart` and symplectically aspherical targets ($E(u) = 0$) reduce to constant spheres via `eq_const_of_totalEnergy_eq_z` (`exists_target_thresholds_of_symplectically_aspherical`). □

Theorem 3.4 (Weitzenböck decomposition (`h_weitz` discharged))**.** *In stereographic coordinates the energy density $e(w)$ of a J-holomorphic sphere is smooth and decomposes as $\Delta e(w) = \mathrm{hessNorm}^2 + \mathrm{curvTerm}$ with $|\mathrm{curvTerm}| \le b\, e(w)^2 + c\, e(w)$ $(c = 0)$. Proved from the centered normal-chart Weitzenböck remainder bound $|\mathrm{rem}| \le |K_{\mathrm{tensor}}(g, J, v)|$ with explicit normal-chart constant $b = (2C_{g,3} + 4C_{J,3}^2 + 8C_{J,3})/c_{\min}^2 \ge 0$ (Theorem 9.3 and Lemma 9.4) and from the Bochner lower bound $\Delta e(w) \ge -b\, e(w)^2$ (Theorem 9.7).*

Proof. At the center of a centered normal chart ($v(0) = 0$) with metric coercivity $c_{\min}\|u\|^2 \le g_0(u, u)$ (`exists_metric_coercive` in Theorem 9.3), `sum_norm_fderiv_sq_le_of_coercive_localEnergyDensi` bounds $\|\partial_s v(0)\|^2 + \|\partial_t v(0)\|^2 \le (2/c_{\min})e(0)$; combining with the metric Hessian contraction (`abs_metricHessianContraction_le`) and almost-complex torsion contraction (`abs_almostComplexTorsionCo`) yields $|K_{\mathrm{tensor}}(g, J, v)| \le ((2C_{g,3} + 4C_{J,3}^2 + 8C_{J,3})/c_{\min}^2)e(0)^2$ (`bound_K_tensor_of_coercive`). Thus $\Delta e(w) = \mathrm{hessNorm}^2 + \mathrm{rem} \ge -|\mathrm{rem}| \ge -b\, e(w)^2$ (`bochner_lower_bound_of_normal_chart_weitz_at_cent`) and `bochner_curvature_decomposition_of_lower_bound` produces the Weitzenböck witnesses. □

Theorem 3.5 (Target monotonicity and small-ball exit (`h_hit` / `h_hit_exit` discharged))**.** *For target-dependent constants $c, r_0 > 0$, $cr^2 \le E(u; u^{-1}B_r(x))$ at boundary-hit radii $r \le r_0$, and nonconstant spheres exit balls of radius $\le r_0$. Proved from the polar coarea radial slicing $F(t) = \int_0^t s(\rho)\, d\rho$ (Lemma 9.4), the Stokes radial differential inequality $F(t) \le \frac{1}{2} t\, F'(t)$ on $[r_{\min}, r]$ integrated via the quotient rule and Mean Value Theorem (`quadratic_energy_growth_of_radial_diff_ineq`), and small-ball exit via stereographic continuity and two-point diameter separation $2r_0 \le \mathrm{dist}(u(z_1), u(z_2))$ (Theorems 9.7 and 9.8).*

Proof. By `hasDerivAt_energy_density_ratio` (Lemma 9.4), the quotient rule gives $\frac{d}{dt}(F(t)/t^2) = (t\, F'(t) - 2F(t))/t^3 \ge 0$ on $[r_{\min}, r]$ whenever $F(t) \le \frac{1}{2} t\, F'(t)$. By the Mean Value Theorem (`Convex.monotoneOn_of_deriv_nonneg`), $t \mapsto F(t)/t^2$ is monotone on $[r_{\min}, r]$, yielding $(F(r_{\min})/r_{\min}^2)r^2 \le F(r)$ (`quadratic_energy_growth_of_radial_diff_ineq`). Combining with the small-radius lower bound $c\, r_{\min}^2 \le F(r_{\min})$ and sublevel inclusion $F(r) \le E(u; u^{-1}B_r(x))$ proves `BoundaryHitMonotonicity` (`boundaryHitMonotonicity_of_radial_diff_ineq`). For `SmallBallExit`, stereographic pullback smoothness plus north-pole continuity gives global continuity of the energy density (`continuous_riemannianEnergyDensity_of_cont_stereo_and_north_pole`), and the triangle inequality $\mathrm{dist}(u(z_1), u(z_2)) \le \mathrm{dist}(u(z_1), x) + \mathrm{dist}(u(z_2), x)$ gives small-ball exit whenever $\mathrm{dist}(u(z_1), u(z_2)) \ge 2r_0$ (`smallBallExit_of_pair_dist_ge`), establishing `HitExitProperty` (`hitExitProperty_of_radial_diff_ineq_and_exit`). □

Theorem 3.6 (Collar oscillation on cylinders (`h_osc` discharged))**.** *For target-dependent constants $C_{\mathrm{osc}}, \varepsilon_{\mathrm{osc}} > 0$, when $E(u; A(r,R)) < \varepsilon_{\mathrm{osc}}$ the oscillation of u on the collar $A(2r, R/2)$ is at most $C_{\mathrm{osc}}\sqrt{E(u; A(r,R))}$. Proved for arbitrary aspect ratios $0 < r < R$ from bounded-modulus ($R \le 18r$) radial/angular slice bounds (Theorem 9.7) and two-sided cylinder energy decay (Theorem 3.7) via dyadic radial chaining (Theorem 9.8) with explicit constants $C_{\mathrm{osc}} = C_{\mathrm{ring}}\sqrt{C_0}\, 3^{a/2} \cdot 2/(1 - 2^{-a/2})$ and $\varepsilon_{\mathrm{osc}} = \min(\varepsilon_{\mathrm{neck}}, \varepsilon_{\mathrm{ring}}/(2C_0))$.*

Proof. On bounded-modulus sub-annuli $r_2 \le 18 r_1$, radial and circular slice bounds combine via the triangle inequality (`satisfiesCylinderCollarOscillation_of_radial_and_angular_bounds`) or collar Lipschitz integration (`satisfiesCylinderCollarOscillationWith_of_lipschitz_collar_ring` and `riemannianEnergyDensity_collar_le_of_disk_regularity`); for general $0 < r < R$, any two collar points $w_1, w_2 \in A(2r, R/2)$ are connected along the dyadic radial chain `dyadicChainPoint` inside parent sub-annuli of aspect ratio ≤ 18, and summing against Theorem 3.7 via `sum_geom_le_inv_one_sub` yields the uniform oscillation bound. □

Theorem 3.7 (Exponential decay on cylinders (`h_decay` discharged))**.** *For target-dependent constants $C_0, a, \varepsilon_{\mathrm{neck}} > 0$, when $E(u; A(r,R)) < \varepsilon_{\mathrm{neck}}$ one has $E(u; A(r_1, r_2)) \le C_0((r/r_1)^a + (r_2/R)^a)E(u; A(r,R))$. Proved from single-step collar Widman hole-filling with explicit constants $C_0 = (1+K)/K > 0$ and $a = \log((1+K)/K)/\log 2 > 0$ (Theorem 9.7), from bounded aspect ratios $R \le M_{\mathrm{rat}} r$, and from `TargetThresholds` (Definition 3.2).*

Proof. Given $r < r_1 < r_2 < R$, choose $k \in \mathbb{N}$ via `exists_nat_pow_near_of_lt_one` so that $2^{-(k+1)} < \max(r/r_1, r_2/R) \le 2^{-k}$, giving $A(r_1, r_2) \subseteq A(2^k r, R/2^k)$ and non-degeneracy $4(2^j r) < R/2^j$ for all $j < k$; disjoint collar-ring superadditivity (`energyMeasure_collar_sum_le_outer_sub_inner`) and iterated Widman contraction (`iterated_dyadic_annulus_energy_le_of_hole_filling`) then bound $E(u; A(r_1, r_2)) \le (K/(1+K))^k E(u; A(r,R)) \le C_0((r/r_1)^a + (r_2/R)^a)E(u; A(r,R))$. □

Theorem 3.8 (Extension across a puncture (`h_ext` discharged))**.** *Every finite-energy punctured J-holomorphic curve $u\colon \mathbb{CP}^1 \setminus \{p\} \to M$ (at either a finite point $p = \varphi_{\mathrm{stereo}}^{-1}(a)$ or $p = \infty$) has vanishing punctured-neighborhood oscillation under Theorem 3.6, admits a unique puncture limit $m_* = \lim_{z \to p} u(z)$ and a unique continuous extension $u_{\mathrm{ext}} = \mathrm{update}(u, p, m_*)$ across p, and satisfies `SatisfiesPunctureSobolevExtension` (Theorem 4.11) once u_{ext} is C^1 with continuous differential at p.*

Proof. By countable intersection of shrinking punctured disks/exteriors (`exists_punctured_disk_energy_lt` and `exists_exterior_annulus_energy_lt`), small-energy collar oscillation (Theorem 3.6) forces vanishing oscillation along $\mathcal{N}_{\neq}(p)$ (`punctured_oscillation_of_local_collar_oscillation`). Completeness of compact M yields a unique limit m_* (`exists_unique_limit_point_of_local_collar_oscillat`) and unique continuous extension $u_{\mathrm{ext}} = \mathrm{update}(u, p, m_*)$ (`exists_unique_continuous_extension_of_local_co`) single-point C^1 and `mfderiv` continuity at p together with global continuity of J (`continuous_J_global`)

and density of $\mathbb{CP}^1 \setminus \{p\}$ (`dense_punctured_sphere`) then establish `WeakJHolomorphicOn` on all of $\mathbb{CP}^1$. □

Theorem 3.9 (Elliptic regularity tower (`h_tower` discharged))**.** *Every weak C^1 J-holomorphic map $u\colon U \subseteq \mathbb{CP}^1 \to M$ is C^n on U for all $n \in \mathbb{N}$ (`RegularityTowerProperty`), proved by induction on n from the base cases $n = 0, 1$ (`WeakJHolomorphicOn.1`) and the local extended-chart regularity step* ContDiffAt $\mathbb{R}$ $(n+1) \Rightarrow$ ContDiffAt $\mathbb{R}$ $(n+2)$ *on* writtenInExtChartAt $I_{\mathbb{CP}^1}$ I z u *(`regularityTowerProperty_of_chart_step`), where the quasilinear Laplacian commutator satisfies $\|N(v, Dv)\| \le (1 + C_{J,0})C_{J,1}\|Dv\|^2$ (`norm_nonlinearLaplacianRHS_le`) and the chart-to-manifold regularity equivalence is discharged from first principles via `contDiffAt_writtenInExtChartAt_of_co` and `contMDiffWithinAt_of_weakJHolomorphicOn_and_contDiffAt_writtenInExtChartAt`.*

Proof. For $n \le 1$, ContMDiffOn $I_{\mathbb{CP}^1}$ I n u U follows directly from $u \in C^1(U, M)$ (`WeakJHolomorphicOn.1.of_l`). For the inductive step $n + 1 \to n + 2$, `contDiffAt_writtenInExtChartAt_of_contMDiffOn` extracts ContDiffAt $\mathbb{R}$ $(n + 1)$ (writtenInExtChartAt $I_{\mathbb{CP}^1}$ I z u) (extChartAt $I_{\mathbb{CP}^1}$ z z) for all $z \in U$ from openness of U, the chart-level elliptic step upgrades this to order $n + 2$, and `contMDiffWithinAt_of_weakJHolomorphicOn_and_contDiffAt_writtenInExtChartAt` converts the extended-chart regularity back to ContMDiffOn $I_{\mathbb{CP}^1}$ I $(n + 2)$ u U. □

Theorem 3.10 (Arzelà–Ascoli smooth extraction (`h_aa` discharged))**.** *For J-holomorphic sequences on an open subset $\Omega \subseteq \mathbb{CP}^1$ under local energy-density bounds, a subsequence converges in $C^\infty_{\mathrm{loc}}(\Omega, M)$ (`ArzelaAscoliProperty`).*

Proof. On a `BoundedChartedTarget`, metric coercivity and radial MVT fencing convert local bounds on `riemannianEnergyDensity` into local Lipschitz bounds (`local_lipschitz_of_bounded_energy_dens`) and equicontinuity on Ω (`equicontinuous_of_local_lipschitz_on_domain`). Diagonalization over a countable dense sequence in Ω (`exists_continuousOn_pointwise_convergent_subseq_on_domain`) extracts a subsequence converging uniformly on compact subsets of Ω to a continuous limit u_{lim}, localizing eventually into fixed target charts (`exists_fixed_chart_ball_eventually_smooth_and_pointwise`). Eventual local chart jet bounds upgrade pointwise chart convergence to all higher jets without passing to a further subsequence (`coreSmoothLocalConvergesOn_of_eventually_local_jet_bounds`), and Faà di Bruno invariance under Möbius frames (`eventuallyHasLocalChartJetBoundsAt_comp_moebius`) combined with discrete frame stabilization yields full frame-invariant `SmoothLocalConvergesOn` (`arzelaAscoliProperty_of_bounded_geometry_and_eventually_jet_bounds`). □

Theorem 3.11 (Smooth extraction off the singular set and defect measure decomposition (`h_conc_extract`))**.** *Along a further subsequence the maps converge in C^∞_{loc} off S to a J-holomorphic u_0, and $\mu = \mu_{u_0} + \sum_{p\in S} \mu\{p\}\delta_p$.*

Proof. Off S, Portmanteau and ε-regularity bound the energy density uniformly on compact subsets (`eventually_gradient_bound_on_compact_southern_hemisphere_of_disjoint_S`). Pointwise convergence of $e(u_{\sigma(\tau(n))}) \to e(u_1)$ under C^∞_{loc} convergence (`tendsto_riemannianEnergyDensity_of_smoot`) Lebesgue dominated convergence against the integrable spherical Jacobian, and monotone Urysohn exhaustion of open subsets of $\mathbb{CP}^1 \setminus S$ yield $\mu_{u_1}(\mathbb{CP}^1) \le \mu(\mathbb{CP}^1) < \top$ and $\mu = \mu_{u_1} + \sum_{p\in S} \mu\{p\}\delta_p$ (`defect_measure_decomp_of_smooth_conv_off_S`). Iterated singularity removal across S by `Finset` induction (`jholomorphic_extension_across_finset`) produces a global J-holomorphic sphere u_0 agreeing with u_1 off S, preserving $\mu_{u_0} = \mu_{u_1}$ (`energyMeasureExtended_congr_off_finset`). □

Definition 3.12 (Genuine defect of a sequence)**.** f is J-holomorphic with bounded finite energy, $\lim_{r\to 0} \lim_n E(f_n; D_r(a)) = m$, and near a the maps eventually stay close to b.

Theorem 3.13 (Realized one-level bubble decomposition (`OneLevelCertificate`, Contract B04))**.** *Every genuine defect of mass $m \ge h$ admits a realized one-level decomposition (subsequence, moving centers and scales, Möbius rescaling frames, frame collapse, C^∞_{loc} convergence off nodes, and genuine child defects).*

Proof. For any genuine defect `IsDefectSeq` of mass $m \ge h$ at a, radial energy continuity and the intermediate value theorem yield a calibrated subsequence σ_0 and positive scales $\lambda_k \to 0$ with $E(f_{\sigma_0(k)}; D_{\lambda_k}(a)) = m - h/2$ (`IsDefectSeq.exists_calibrated_subseq_scales`), while compactness of $K \subset \mathbb{CP}^1 \setminus \{\infty\}$ and $K \subset \mathbb{CP}^1 \setminus \{a\}$ gives uniform forward and inverse affine-frame collapse (`affine_scaling_tendstoUniformlyOn_punctured_north_pole`, `affine_scaling_inv_tendstoUniforml` assembling `SeqOneLevelDecomposition.of_rescaled_limit` and `oneLevelCertificate_of_rescaled_limit`. □

Theorem 3.14 (Concentration atoms as genuine defects (`ConcDefectCertificate`))**.** *Every atom of the concentration measure is a genuine defect of a rotated subsequence of $u_{\sigma(n)}$, with its true mass and base value $u_0(p)$.*

Proof. For any finite atom $p = \varphi^{-1}_{\mathrm{stereo}}(a) \in S$, isolating disks (`exists_isolating_disk_radius`), null circle boundaries (`frontier_disk_at_null_of_isolating`), Portmanteau disk convergence (`isolating_disk_defect_limit_of_decomp`), and radial continuity of $E(u_0; D_r(a)) \to 0$ unconditionally yield $\lim_{r\to 0} \lim_n E(u_{\sigma(n)}; D_r(a)) = m_p$ and ghost-root total energy convergence

(defect_mass_limit_of_concentration_finite_point). On the open southern hemisphere $\|a\| < 1$, compact-annulus ε-regularity gradient bounds (eventually_gradient_bound_on_compact_southern_h Lipschitz equicontinuity (local_lipschitz_of_bounded_energy_density), and pointwise convergence off S upgrade to uniform convergence on $A(a; r/2, r)$ (tendstoUniformlyOn_of_pointwise_and_equico proving IsDefectSeq unconditionally (isDefectSeq_of_concentration_southern_hemisphere). Pulling concentration data back along a conformal Möbius rotation sending $0 \mapsto p$ (weakMeasureConvergence_m reduces every atom $p \in S$ to $a = 0$ (concDefectCertificate_of_rotations). □

Theorem 3.15 (Proved extension theorems: Möbius 3-point transitivity & rigidity, $\mathrm{PSL}(2,\mathbb{C})$ finiteness, sequential metrizability, and marked/varying-J extraction)**.** *Every matrix in* $\mathrm{GL}(2,\mathbb{C})$ *lifts to a smooth* MoebiusFrame *on* $\mathbb{CP}^1$ *(MoebiusFrame.ofMatrix), discharging 3-point homeomorphism rigidity (moebiusFrame_three_point_rigidity) and 3-point transitivity (exists_moebiusFrame_thr unconditionally; quotienting by scalar matrices yields finite effective component symmetries and Kontsevich's stability equivalence from isolated fibers (stability_iff_finite_effective_automorphisms_of_f sequential metric compatibility proves metrizability of the Gromov topology (moduli_space_metrizable_of_sequ and core/marked/varying-J limit extraction holds with universal energy and component bounds (gromov_compactness_varying_J_of_witness_and_thresholds, marked_gromov_compactness_of_rotations_ moduli_space_three_lt_eps_sph_compact_hausdorff_of_chart_cont).*

Proof. Orthogonal $\mathbb{R}^3$ sphere isometries realize unit rotations and $w \mapsto 4/w$ smoothly on $\mathbb{CP}^1$, generating $\mathrm{GL}(2,\mathbb{C})$ by Bruhat decomposition (MoebiusFrame.ofMatrix). Projective 3-point uniqueness lifts along cp1ToOnePoint_injective to give moebiusFrame_three_point_rigidity and ghost_component_pure_automorphism_trivial_uncond; in $\mathrm{PSL}(2,\mathbb{C}) =$ EffectiveMoebiusFrame, the identity-homeomorphism fiber is a subsingleton and the Intermediate Value Theorem produces three distinct values on any nonconstant component, injecting its symmetry group into a finite triple of point fibers (effective_principal_symmetries_finite_of_finite_fibers). Any metric satisfying McDuff–Salamon's sequential axioms (MS1)–(MS2) has open sets identical to moduliSpaceTopology (moduli_space_metrizable_of_sequential_metric), establishing $\overline{\mathcal{M}}_{0,3}(M, J, \leq B) \cong M$ compact Hausdorff for $0 \leq B < \varepsilon_{\mathrm{sph}}$ (moduli_space_three_lt_eps_sph_compact_hausdo Finite-product compactness on $V \times \mathrm{Fin}\, k \to \mathbb{CP}^1$ extracts convergent pulled-back marks along core bubble trees (exists_convergent_pulled_back_marks_subseq), while operator-norm Cauchy–Riemann closure and TargetThresholds yield varying-J component bounds (gromov_compactness_varying_J_ □

Chapter 4

Local analysis

Lemma 4.1 (Scalar mean-value inequality (A01)). *If $e \geq 0$ is C^2 with $\Delta e \geq -be^2$ on D_r and $\int_{D_r} e < \eta$, then $\sup_{D_{r/2}} e \leq Cr^{-2} \int_{D_r} e$.*

Proof. Heinz-type argument in flat coordinates. □

Lemma 4.2 (Bochner inequality (A02)). *$\Delta e \geq -be^2 - ce$ for the energy density of a J-holomorphic sphere.*

Proof. Algebra from Theorem 3.4. □

Theorem 4.3 (ε-regularity (A03)). *Small energy on a disk bounds the energy density on the half disk.*

Proof. Mean-value inequality applied to the Bochner inequality. □

Lemma 4.4 (Flat elliptic estimates (A04)). *Interior Cauchy–Riemann estimates, 2D Morrey–Sobolev embedding, Cauchy–Pompeiu Hölder gain and one Hölder bootstrap step, in flat coordinates.*

Proof. Mathlib's Cauchy integral formula and a near/far splitting. □

Theorem 4.5 (Nonlinear bootstrap (A05)). *Weak J-holomorphic maps are smooth.*

Proof. Immediate from Theorem 3.9. □

Theorem 4.6 (Local smooth compactness (A06)). *Under local gradient bounds a subsequence converges in C^∞_{loc} to a J-holomorphic limit.*

Proof. Extraction from Theorem 3.10; limit is J-holomorphic by Lemma 2.12. □

Theorem 4.7 (Target monotonicity (A08)). *$cr^2 \leq E(u; u^{-1}B_r(x))$ for small r when u leaves $B_r(x)$.*

Proof. From Theorem 3.5. □

Theorem 4.8 (Sphere energy gap (A07)). *Nonconstant J-holomorphic spheres have energy at least $\varepsilon_{\mathrm{sph}} > 0$.*

Proof. Monotonicity at a ball the sphere must exit. □

Theorem 4.9 (Cylinder localization (A09)). *Small annular energy confines the collar image to a small ball.*

Proof. From Theorem 3.6. □

Theorem 4.10 (Exponential decay on cylinders (A10)). *Annular energy decays exponentially away from the collars.*

Proof. From Theorems 3.7 and 9.7. □

Theorem 4.11 (Extension across a puncture (A11)). *A finite-energy punctured J-holomorphic sphere extends continuously and weakly J-holomorphically.*

Proof. From Theorem 3.8. □

Theorem 4.12 (Removal of singularities (A12)). *The extension is a unique smooth J-holomorphic sphere.*

Proof. Bootstrap of the extension; uniqueness by density of the punctured sphere. □

Theorem 4.13 (Thresholds from analytic certificates (B01)). *Theorems 3.4, 3.5 and 3.7 yield a threshold bundle with $0 < h \leq \min(\varepsilon_{\mathrm{reg}}/2, \varepsilon_{\mathrm{sph}}, \varepsilon_{\mathrm{neck}}/2)$.*

Proof. Take minima of the constants. □

Chapter 5

Concentration and bubbling

Theorem 5.1 (Prokhorov compactness on $\mathbb{CP}^1$)**.** *Bounded sequences of finite measures on $\mathbb{CP}^1$ have weakly convergent subsequences.*

Proof. Diagonal argument over a dense sequence in $C(\mathbb{CP}^1)$ and Mathlib's `RealRMK.rieszMeasure`. □

Theorem 5.2 (Weak compactness of energy measures (C01))**.** *Energy measures of a bounded-energy sequence have a weakly convergent subsequence.*

Proof. From Theorem 5.1 and Theorem 3.1. □

Definition 5.3 (Concentration data)**.** A subsequence, a limit measure μ, a finite singular set S and a J-holomorphic limit u_0 off S.

Theorem 5.4 (Existence of concentration data (C02))**.** *Every bounded-energy sequence admits concentration data.*

Proof. Weak compactness, finiteness of S, and Theorem 3.11. □

Theorem 5.5 (Defect identity (C03))**.** $\mu = \mu_{u_0} + \sum_{p \in S} m_p \delta_p$ *with* $m_p \geq \varepsilon_{\mathrm{sph}}$*, and* $E(u_{\sigma(n)}) \to E(u_0) + \sum m_p$*.*

Proof. Finite-measure Portmanteau. □

Lemma 5.6 (Calibrated rescaling (B02))**.** *Hofer's lemma specialized to the plane gives centers and scales with $E(f_n; D_{\lambda_n}(a_n)) = m - h/2$.*

Proof. From `Mathlib.Analysis.Hofer`. □

Theorem 5.7 (Neck vanishing (B03))**.** *Neck energy and neck diameter tend to zero with the stated quantifier order.*

Proof. From cylinder localization and Theorem 3.6. □

Definition 5.8 (One-level decomposition)**.** A bubble v, child points q and masses with $m = E(v) + \sum m_i$, $v(\infty) = b$, and calibration.

Lemma 5.9 (Ghost branching (B05))**.** *A constant bubble has at least two children.*

Proof. Calibration arithmetic. □

Lemma 5.10 (Strict rank decrease (B06))**.** $\lfloor m_i/h \rfloor < \lfloor m/h \rfloor$*; well-founded induction on this rank.*

Proof. Children have mass at most $m - h$. □

Definition 5.11 (Local and realized defect trees)**.** A finite stable tree of J-holomorphic spheres of total energy m attached at b; a realized local defect tree also records the subsequence, per-vertex Möbius frames, transition collapse and C^∞_{loc} convergence off incident nodes.

Theorem 5.12 (Resolution of a genuine defect (B07))**.** *Every genuine defect of mass $m > 0$ is resolved by a realized local defect tree.*

Proof. Energy-rank induction; realized children are grafted sequentially along composed subsequences and telescoped frames (`graft_child_defect_trees_realized`). □

Chapter 6

Assembly and the master theorem

Lemma 6.1 (Global frames (G01))**.** *Rescaling data on a rooted tree determine global Möbius frames with the transition relations.*

Proof. Well-founded recursion on depth. □

Definition 6.2 (Grafted tree and prestable limit data)**.** A tree rooted at u_0 with defect trees attached at S; realized grafted trees and prestable limit data carry the subsequence, Möbius frames, transition collapse and C^∞_{loc} convergence.

Theorem 6.3 (Realized root-sphere grafting theorem)**.** *Grafting realized local defect trees onto the root curve u_0 across all concentration atoms $p \in S$ produces a realized grafted tree with global frame collapse and C^∞_{loc} convergence off nodes.*

Proof. Finite-set induction over $s \subseteq S$, composing subsequences and rotating/telescoping child frames. □

Theorem 6.4 (Prestable limit (G01))**.** *Every bounded-energy sequence of J-holomorphic spheres has prestable limit data.*

Proof. Concentration data, defect identity, realized B07 resolution at each atom of S via Theorems 3.13 and 3.14, and Theorem 6.3. □

Theorem 6.5 (Stabilization (G02))**.** *Unstable ghost roots can be pruned or suppressed, yielding a stable limit.*

Proof. Strong induction on $|V|$. □

Theorem 6.6 (Core compactness (G03))**.** *A subsequence converges to a stable sphere tree limit.*

Proof. Assembly followed by stabilization. □

Theorem 6.7 (Genus-zero Gromov compactness)**.** *Every bounded-energy sequence of nonconstant J-holomorphic spheres admits a subsequence converging to a stable sphere tree limit with universal principal, ghost, and vertex bounds $Ph \le B$, $G+2 \le P$ and $|V|+2 \le 2P$ (for $|V| > 1$), and $|V| \le \max(1, 2\lfloor B/h \rfloor - 2)$.*

Proof. Construct the threshold bundle $h > 0$ via Theorem 3.3 (`TargetThresholds.of_properties` / `TargetThresholds.of_bounded_geometry_certs`), apply core compactness Theorem 6.6 (`gromov_compactnes`) and bound principal and ghost components via Theorem 2.15 (`component_count_bounds`). □

Chapter 7

Extensions

Theorem 7.1 (Varying almost-complex structures (E01))**.** *Limits of J_n-holomorphic maps with $J_n \to J$ are J-holomorphic on open domains, and varying-J bubble-tree limits satisfy $\sum_v E(u_v) \le B$ and universal component bounds.*

Proof. Operator-norm squeeze on $J_n(u_n(z))\,du_n(z) \to J(u(z))\,du(z)$ (`jholomorphic_limit_of_varying_J`, `varying_J_limit_jhol_on_open`) combined with uniform and limit `TargetThresholds` (`gromov_compactness_v` from Theorem 3.15. □

Theorem 7.2 (Marked compactness (E02))**.** *Compactness for k-marked J-holomorphic spheres, with $G \le P + k - 2$ and $|V| \le \max(1, 2\lfloor B/h \rfloor + k - 2)$.*

Proof. Constant 3-marked spheres normalize $(0, 1, \infty) \mapsto (z_{n,0}, z_{n,1}, z_{n,2})$ via 3-point Möbius transitivity (`marked_gromov_compactness_three_const_of_chart_cont`), while nonconstant k-marked spheres combine Theorem 6.6 with unconditional finite-product mark extraction (`exists_convergent_p` `marked_gromov_compactness_of_rotations_nodal_disjoint`) from Theorem 3.15. □

Theorem 7.3 (Automorphisms (E04))**.** *Unconditional Möbius 3-point homeomorphism rigidity, triviality of pure automorphisms on stable ghost components, and finiteness of* $\mathrm{Aut}(T, \{u_v\}, \{z_{vw}\})$ *modulo scalar matrices.*

Proof. Three-point homeomorphism rigidity (`moebiusFrame_three_point_rigidity`) forces every ghost pure automorphism to be $\mathrm{id}_{\mathbb{CP}^1}$ (`ghost_component_pure_automorphism_trivial_uncond`), and isolated principal fibers give finiteness of `EffectiveStableMapAutomorphism` (`stability_iff_finite_effe` via Theorem 3.15. □

Theorem 7.4 (Moduli space (E05))**.** *Sequential metrizability and compact Hausdorff structure of $\overline{\mathcal{M}}_{0,k}(M, J, \le B)$ and $\overline{\mathcal{M}}_{0,k}(M, J, A)$, with unconditional $\overline{\mathcal{M}}_{0,3}(M, J, \le B) \cong M$ for $0 \le B < \varepsilon_{\mathrm{sph}}$.*

Proof. Sequential metric compatibility implies topological agreement with `moduliSpaceTopology` (`moduli_space_metrizable_of_sequential_metric`), yielding Hausdorff separation and compactness (`moduli_space_compact_hausdorff_of_sequential_metric`) and $\overline{\mathcal{M}}_{0,3}(M, J, \le B) \cong M$ (`moduli_space_three_lt_eps_sph_compact_hausdorff_of_chart_cont`) via Theorem 3.15. □

Chapter 8

Certificate audit

Remark 8.1 (Withdrawn certificates and definition repairs). Up to commit `0a3de26` the master theorem assumed `h_one_level` (a decomposition for *every* mass $m \geq h$) and `h_graft_extract` (convergence to *every* grafted tree with arbitrary masses). `tests/CertificateSanity.lean` proves: S2, the former forces a sphere of energy exactly m for each $m \in [h, 2h)$ and so fails on any target missing one such value (e.g. $\mathbb{CP}^1$); S4, the latter is contradictory for sequences with convergent energies and two admissible trees. S1 checks that $u_n = \mathrm{id}_{\mathbb{CP}^1}$ satisfies the non-certificate hypotheses, S3 that Definition 3.12 fails without energy concentration, S5 that `affine_scaling` genuinely translates by its center $a \in \mathbb{C}$, S6 that `SmoothLocalConvergesOn` holds for the nonconstant identity map on $\mathbb{CP}^1$, S7 (`s7_old_h_aa_prop_false_without_jhol`) that unconditioned Arzelà–Ascoli smooth extraction fails without J-holomorphicity, and S8 (`s8_old_h_decay_prop_false_without_smallness`) that annular energy decay fails without an energy-smallness threshold $\varepsilon_{\mathrm{neck}} > 0$. Both over-strong certificates were replaced by Theorems 3.13 and 3.14, and `GraftCertificate` was eliminated in favour of the proved theorem Theorem 6.3.

Chapter 9

Normal-chart infrastructure (N1–N3)

Theorem 9.1 (Finite good atlas and `mfderiv` factorization (N1))**.** *A compact manifold has a finite atlas with a positive Lebesgue radius, and* ***`mfderiv`*** *factors through its charts via* ***`tangentCoordChange`****.*

Proof. Finite subcover of precompact coordinate balls and Lebesgue number lemma; chain rule for `extChartAt` and `tangentCoordChange_comp`. □

Theorem 9.2 (Energy density certificate and unconditional `h_fin` discharge (N1))**.** *Every C^1 map $u : \mathbb{CP}^1 \to M$ on a* ***`BoundedChartedTarget`*** *is globally Lipschitz on $\mathbb{CP}^1$, has uniformly bounded manifold differential $\|du\|$ and Riemannian energy density $e(u) \le C$, unconditionally inhabits* ***`EnergyDensityCertificate`****, and satisfies $\mu_u(\mathbb{CP}^1) < \top$ (Theorem 3.1).*

Proof. Local Lipschitzness from `ContDiffAt.exists_lipschitzOnWith` and chart bi-Lipschitz bounds, compactness of $\mathbb{CP}^1$ (`LocallyLipschitzOn.exists_lipschitzOnWith_of_compact`), `norm_fderiv_le_of_lip'`, and comparison with the finite area of $\mathbb{CP}^1$ under the spherical Jacobian. □

Theorem 9.3 (Quadratic Taylor normal charts, Koszul cancellation and general-manifold construction (N2))**.** *Quadratic Taylor normalization $\Phi_{y_0}(v) = y_0 + v - \frac{1}{2}\Gamma_{y_0}(v, v)$ yields local diffeomorphisms via the inverse function theorem; composing with atlas charts produces centered charts Ψ_p with $\Psi_p(0) = p$, $d\Psi_p(0) = \mathrm{id}$, and algebraic Koszul cancellation of the first pulled-back metric derivatives at 0 (****`pulledBackMetricAt`****). **`UniformNormalChartFamily`** packages the 15 geometric fields (uniform radius, bi-Lipschitz distance comparison, curvature and C^3 tensor bounds), is inhabited on $(\mathbb{CP}^1, \omega_{\mathrm{FS}}, J_{\mathbb{CP}^1})$ via* ***`concrete_CP1`*** *and* ***`instBoundedChartedTargetCP1`****, and is constructed unconditionally on any finite-dimensional* ***`BoundedChartedTarget`*** *via* ***`exists_metric_coercive`****,* ***`christoffelFromKoszul`****, and* ***`construct_uniform_normal_chart_family`****.*

Proof. `ContDiffAt.toOpenPartialHomeomorph`, Koszul symmetry cancellation, unit-sphere metric coercivity, 3rd-order Faà di Bruno bounds (`norm_iteratedFDeriv_comp_taylor_le`), and explicit stereographic bi-Lipschitz bounds on $\mathbb{CP}^1$. □

Lemma 9.4 (Coordinate Bochner–Weitzenböck and radial monotonicity building blocks)**.** *At the center of a normal chart, vanishing of first metric derivatives implies vanishing of the Koszul form and Christoffel symbols and quadratic energy-density control* $|K_{\mathrm{tensor}}| \le (2C_{g,3} + 4C_{J,3}^2 + 8C_{J,3})e(0)^2$*, while any lower bound* $\Delta e(w) \ge -be(w)^2$ *splits algebraically into* $\Delta e(w) = \mathrm{hessNorm}^2 + \mathrm{curvTerm}$ *with* $|\mathrm{curvTerm}| \le be(w)^2$*. In polar coordinates, disk energy splits into circle slices with* $F'(r) = \int_0^{2\pi} e(r,\theta) r\, d\theta$*, and* $F(t) \le \frac{1}{2} tF'(t)$ *integrates to* $(F(r_{\min})/r_{\min}^2)r^2 \le F(r)$*.*

Proof. Operator-norm bounds on `iteratedFDeriv`, polar integration, and `monotoneOn_of_deriv_nonneg`. □

Lemma 9.5 (Coordinate singularity removal and elliptic bootstrap building blocks)**.** *Iterated Widman hole-filling contraction gives geometric annular decay; finite punctured-sphere energy implies vanishing annular energy tail, vanishing collar oscillation (under* `h_osc`*), and Cauchy filter convergence across the puncture on compact* M*; the quasilinear Laplacian commutator satisfies* $\|N(v, Dv)\| \le (1+C_{J,0})C_{J,1}\|Dv\|^2$*; and local Lipschitz gradient bounds yield equicontinuity and separable dense-set Arzelà–Ascoli extraction on subdomains.*

Proof. Dyadic induction, `CompleteSpace.complete` on M, operator-norm bounds, and `tendstoUniformlyOn_o` □

Theorem 9.6 (Phase 1: Unconditional C^0 puncture extension and Möbius-frame Arzelà–Ascoli jet upgrade)**.** *(1) From finite punctured-sphere energy and collar oscillation (*`h_osc`*) alone, any two points in a punctured disk lie in a single collar of energy* $< (\varepsilon/5)^2$*, yielding vanishing punctured-disk oscillation, unconditional existence of the puncture limit* $m_* \in M$*, unconditional continuous extension* $u_{\mathrm{ext}} = \mathrm{update}\, u\, p\, m_*$*, and global* `WeakJHolomorphicOn` *extension across* p *by punctured-sphere density whenever* u_{ext} *is* C^1 *with continuous Cauchy–Riemann defect at* p*. (2) Under discrete frame topology, any finite list of* `ConvergingFrames` *eventually acts by a single composite* $\phi \in \mathrm{MoebiusFrame}$*, reducing* `SmoothLocalConvergesOn` *to* `CoreSmoothLocalConvergesOn` *under fixed Möbius frames, while* `tendsto_iteratedFDeriv_writtenInExtChar` *upgrades pointwise chart convergence and local jet bounds to* C^∞ *smoothness of the limit and convergence of all higher jets* `iteratedFDeriv` $\mathbb{R}$ k *without passing to a further subsequence.*

Proof. Single-collar placement $r = \frac{1}{4}\min(|w_1 - a|, |w_2 - a|)$, `dense_punctured_sphere`, `tendsto_discrete`, and `tendstoUniformlyOn_iteratedFDeriv_of_pointwise_of_local_jet_bounds`. □

Theorem 9.7 (Phase 2: Conditional reductions for Bochner, monotonicity, collar oscillation and annular decay)**.** *(1) For Certificate 1 (*`h_weitz`*),* `bochner_curvature_decomposition_of_lower_bound` *converts any pointwise Bochner lower bound* $\Delta e(w) \ge -be(w)^2$ *into global stereographic Weitzenböck witnesses* hessNorm *and* curvTerm *satisfying* $\Delta e(w) = \mathrm{hessNorm}^2 + \mathrm{curvTerm}$ *and* $|\mathrm{curvTerm}| \le be(w)^2$*. (2) For Certificate 2 (*`h_hit_exit`*),* `sublevel_energy_lower_bound_of_radial_diff_ineq` *and* `boundary_hit_energy_of_radial_diff_ineq` *integrate* $F(t) \le \frac{1}{2}tF'(t)$ *from an infinitesimal scale* $r_{\min} \in (0, r]$ *to obtain* $c_0 r^2 \le E(u; u^{-1}B_r(x))$ *and the boundary-hit disjunction. (3) For Certificates 3–4 (*`h_osc` *and* `h_decay`*),* `satisfiesCylinderCollarOscillation_of_radial_and_angular_bound` *combines radial segment and circular slice bounds* $\le 2\sqrt{E}$ *into* $\operatorname{osc} u \le 4\sqrt{E(u; A(r, R))}$*, while* `iterated_dyadic_annulus_energy_le_of_hole_filling` *and* `satisfiesCylinderEnergyDecayWith_of_colla`

iterate Widman's hole-filling step across nested dyadic annuli $A(2^j r, R/2^j)$ and interpolate continuous sub-annuli $r < r_1 < r_2 < R$ to yield continuous two-sided power decay with $C_0 = (1+K)/K$ and $a = \log((1+K)/K)/\log 2$.

Proof. `decomposition_of_bochner_lower_bound`, `quadratic_energy_growth_of_radial_diff_ineq`, triangle inequality on collar slices, and `le_pow_hole_filling_of_step`. □

Theorem 9.8 (Step 1–2 first-principles reductions: Lipschitz equicontinuity, punctured oscillation, bounded-modulus dyadic chaining and zero-energy constancy)**.** *(1) `local_lipschitz_of_bounded_energy` discharges `h_aa_lip` from first principles by combining the Riemannian energy density bound (`norm_mfderiv_le_of_coercive`), chart bi-Lipschitz control, and radial line-segment integration (`image_le_of_liminf_slope_right_le_deriv_boundary`). (2) `punctured_oscillation_of_local_collar_osc` discharges `h_punc_osc` for every $p \in \mathbb{CP}^1$ (handling $p =$ north_pole via exhausting outer annuli in `punctured_north_pole_oscillation_of_collar_oscillation`) from local collar oscillation. (3) `local_collar_oscillation_of_bounded_modulus_and_decay` and `collarOscillationProperty_of_bound` reduce arbitrary-aspect-ratio collar oscillation ($r < R$) to bounded-modulus rings ($R \le 18r$) plus two-sided cylinder energy decay by chaining dyadic rays (`dist_le_sum_dyadic_chain_bounded_modulus_and_dec`) and summing the geometric series $\sum_{k<N}(2^{-a/2})^k \le 1/(1-2^{-a/2})$. (4) `eq_const_of_totalEnergy_eq_zero` and `totalEnergy_pos_of_nonconstant` prove from first principles that zero-energy C^1 spheres are constant and every nonconstant C^1 sphere has strictly positive energy $E(u) > 0$.*

Proof. Metric coercivity, radial MVT fencing, outer-annulus countable additivity at north_pole, dyadic ring chaining with `sum_geom_le_inv_one_sub`, and `IsLocallyConstant.apply_eq_of_preconnectedSpac` □

Theorem 9.9 (Phase 3: Conditional elliptic regularity tower and certificate constructor)**.** *`contMDiffOn_infty_of_weakJHolomorphicOn_and_step` assembles a given inductive regularity step $C^k \to C^{k+1}$ into C^n and C^∞ regularity and smooth J-holomorphicity (`jholomorphicOn_of_weakJHolomorp`). `UniformNormalChartCertificates.of_first_principles` packages the Phase 1–3 and first-principles reductions into `UniformNormalChartCertificates` when supplied with the remaining local coordinate PDE hypotheses (with `h_aa_lip` and `h_punc_osc` discharged unconditionally and `h_osc_slices` reduced to bounded-modulus rings $R \le 18r$).*

Proof. Induction via `contMDiffOn_nat_of_weakJHolomorphicOn_aux`, `nonlinear_elliptic_bootstrap_of_tow` and synthesis of Phases 1–3. □

Theorem 9.10 (Local-chart certificate bridge from normal chart certificates (N3))**.** *Given a `UniformNormalChartFamily` and conditioned `UniformNormalChartCertificates` (where oscillation, decay, puncture removal and Arzelà–Ascoli extraction are conditioned on J-holomorphicity and smallness/non-degenerate radii, and proved inhabited on constant curves via `conditioned_certificates_i`*

the bridge theorems instantiate Theorems 3.4 to 3.10. Unconditional derivation of `UniformNormalChartCertifi` *from* `ClosedTameTarget` *for general nonconstant curves remains open.*

Proof. Projections, degenerate-collar splitting, and curvature constant combination. □